Admission to High Performance Geo-Computing Research Group

1. Introduction to School and Institute

1.1 About SIGS

Launched in March of 2019, Tsinghua Shenzhen International Graduate School (SIGS) is a research and graduate education institution of Tsinghua University located in southern China. Building upon Tsinghua's academic legacy and Shenzhen's innovative resources, our students and faculty are dedicated to tackling global challenges through cutting-edge research and collaboration. Tsinghua SIGS has set up academic disciplines that promote interdisciplinary research and learning. They include Materials Science, Data Science & Information Technology, Biopharmaceutical & Health Engineering, Ocean Engineering, Future Human Habitats, Environment & Ecology and Innovation Management.

1.2 About iDI

The Institute of Data and Information (iDI) adheres to the principles of innovation, crossover, integration, industrialization, and internationalization, serves and supports the needs of major national projects and development plans, and is oriented to the industrial demands and applications of the Guangdong-Hong Kong-Macao Greater Bay Area. Relying on the top-tier engineering education system of Tsinghua

University, iDI cooperates with RWTH Aachen University, Technical University of Munich and etc. to attract top international talents, and strives to build a world-class highland for talents in data and information-related fields, a high-level postgraduate training base, and a bridgehead in the Guangdong-Hong Kong-Macao Greater Bay Area for the incubation and technology transfer of innovative achievements.

2. High Performance Geo-Computing Research Group

2.1 About HPGC

High Performance Geo-Computing (HPGC) research group focuses on designing highly efficient and scalable simulation and AI applications that leverage multi-core, many-core, and reconfigurable technologies.

We work at the intersection of high-performance computing, computational science, and emerging heterogeneous architectures, tackling grand challenges in atmospheric modeling, earthquake simulation, quantum computing, and more.

We have a strong track record of groundbreaking research, including contributions to extreme-scale simulations on some of the world’s fastest supercomputers. Our team has been recognized with prestigious awards, including the ACM Gordon Bell Prize (three times) and best paper awards at top conferences.

Our group members have published 200+ outstanding achievements in important journals and academic conferences in the fields of geology and computer science, such as Nature, Super Computing, Remote Sensing of Environment, CVPR, RSE, ISPRS, etc.

And we have undertaken a number of projects such as the key Research and development plan of the Ministry of Science and Technology, the National Natural Science Foundation of China, and have cooperated with Huawei, Microsoft, IBM, Intel, Norway Oil, Aramco, CNPC, Schlumberger and many other enterprises.

Group Website：https://www.thuhpgc.net

2.2 Group Leader

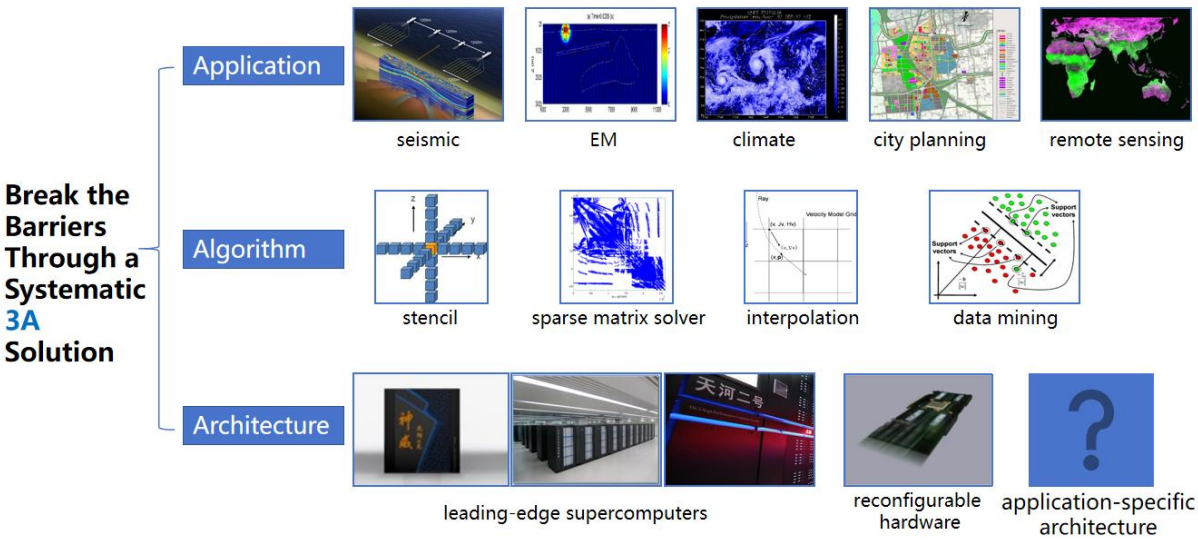
Haohuan Fu, Professor of Tsinghua University, focused on cross-integration research of High Performance Computing, AI and Earth system science, and has made significant contributions in scaling over tens of millions of heterogeneous cores for a series of key applications (e.g. climate model, earthquake simulation, etc.). Professor Fu has published more than 200 journal and conference papers, and have led multiple key research and development programs.

Professor Fu won three ACM Gordon Bell Prizes, the highest award in the field of high performance computing (fully-implicit atmospheric dynamic solver in 2016, non-linear earthquake simulation in 2017, and large-scale quantum circuit simulation in 2021), also won 2016 Tsinghua-Inspur Computational Earth Science Young Researcher Award, 2017 Chinese Institute of Electronics Award For Science and Technology Advancements (Only 14 worldwide), 2020 China Youth May Fourth Medal, 2022 Tsinghua University: Team of Excellence , 2022 CCF HPC China: The Best HPC Application Nomination, and 2023 Wu Wenjun Award for AI Science and Technology Advancements (Second Class).

2.3 Our research interests

The name of our group describes our major research interest, which is to motivate the interaction between High Performance Computing (HPC) and geoscience. The goal is to bridge the gap between the grand challenges in earth science, and the emerging computing methods and hardware platforms.

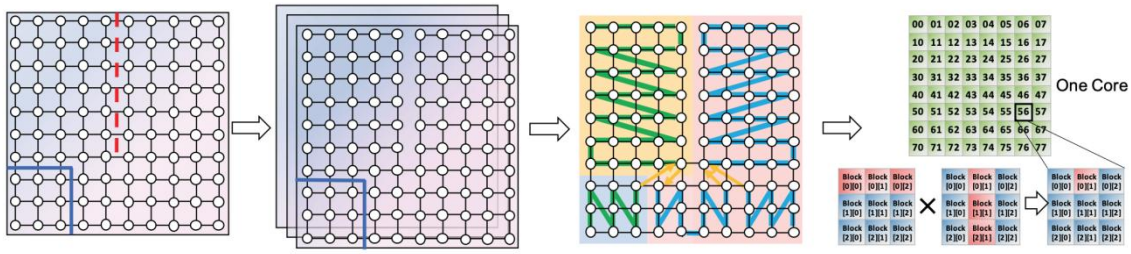
Now, we are focusing on **"HPC + AI + X"** research, specifically conducting cross disciplinary research through **Architecture, Algorithm and Application**!



2.3.1 Physics-based numerical simulation on supercomputers with tens of milliions of cores

Scalable, convenient, and efficient mapping of physics-based numerical models (e.g. climate, earthquake, quantum circuits, etc.) to 100-peta or even exa-scale supercomputers, with tens of millions of cores. By providing parallel algorithms, code migration and performance tuning tools, I strive for better solutions of more challenging scientific problems on faster supercomputers.

Representative paper:

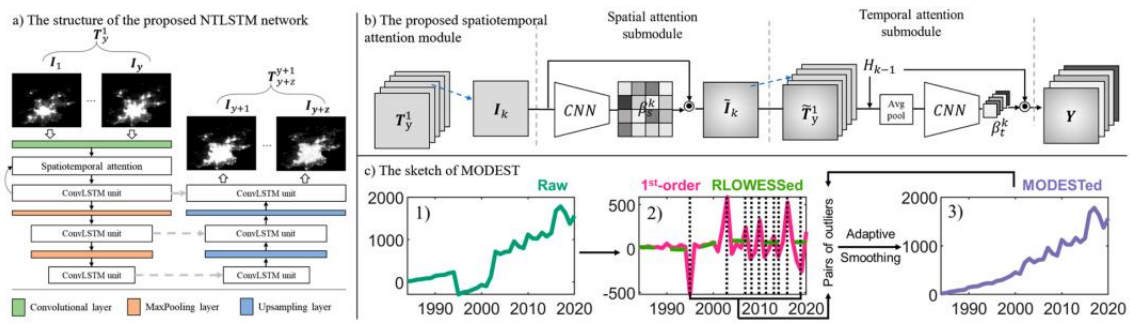


Yong Liu, Xin Liu, Fang Li, Haohuan Fu, etc., **Closing the" quantum supremacy" gap: achieving real-time simulation of a random quantum circuit using a new Sunway supercomputer**, Proceedings of the International Conference for High Performance Computing, 2021(ACM Gordon Bell Prize).

2.3.2 High performance automated processing and analysis platform for large spatial and temporal scale observation data

Large-scale automated data processing methods that make utilization of all available data for the solving process of a particular scientific problem. By minimizing the manual efforts involved, we works on data compression, fusion, highly-scalable weakly-supervised or unsupervised machine learning methods, as well as data-driven multi-modal models, so as to enable the data analysis and understanding of the Earth system (or other scientific scenarios).

Representative paper:

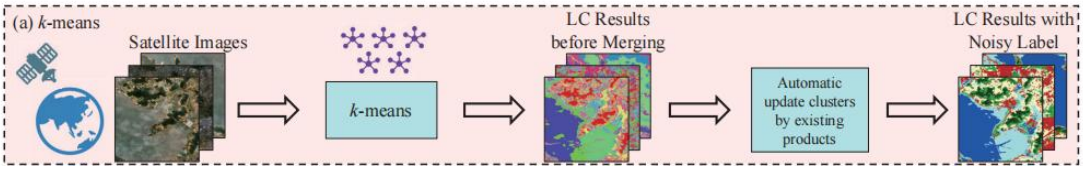


Lixian Zhang, Zhehao Ren, Bin Chen, Peng Gong, Bing Xu, Haohuan Fu*, **A Prolonged Artificial Nighttime-light Dataset of China (1984-2020)**, Scientific Data, 2024.

2.3.3 Intelligent processing and interpretation of geological big data driven by artificial intelligence methods

We also focus on remote sensing and intelligent interpretation method based on multi-source data fusion, high precision weather and climate prediction method based on deep learning. We believe such methods would lead to potential solutions of some of the most grand scientific challenges, such as accurate prediction of raindrop in three-month time, or detailed simulation of molecular dynamics to a scale that finally bridges the physical scenario in the lab and the numerical scenario in a computer.

Representative paper:



Yi Zhao, Juepeng Zheng, Haohuan Fu*, etc., **SW-LCM: A Scalable and Weakly-supervised Land Cover Mapping Method on a New Sunway Supercomputer**, 37th IEEE International Parallel and Distributed Processing Symposium, 2023.

We warmly welcome students who are interested in the above research directions or other related subjects to join us!

2.4 Career of our graduates

In recent years, HPGC has trained over 3 postdocs, 10 master's students and over 10 doctoral students. The following photo ,which shows the career path of some graduates, was taken in 2018, and all the students in the photo have already graduated. Most of our graduates are engaged in teaching and research work in universities or

scientific research work in enterprises.

Group Photo of 2018

First Row, From left to right

1. Liu, Cewen: PhD 2024; Stanford University, Postdoc
2. Zheng, Weijie: PhD 2018; Harbin Institute of Technology(Shenzhen), Associate Professor
3. Fu, Haohuan
4. Zhang, He: Master 2018; China Petrochemical Corporation, Researcher
5. Chen, Binwei: PhD 2020; Sangfor Technology Co., Ltd, Engineer
6. Li, Weijia: PhD 2019; Sun Yat sen University, Associate Professor

Back Row, From left to right

1. Dong, Runmin: PhD 2022; Tsinghua, Shuimu Scholar, Postdoc
2. Xia, Maocai: Master 2019; SenseTime Technology Co., Ltd, Senior Algorithm Engineer
3. Wang, Xin: PhD 2024; National University of Singapore, Postdoc
4. Wei, Tengpeng: Master 2014; JPMorgan, Researcher
5. You, Yang: Master 2015, National University of Singapore, President Young Professor
6. Li, Ouyi: Master 2019, ByteDance, Engineer
7. Xu, Jingheng: PhD 2019, Sangfor Technology Co., Ltd, Engineer
8. Shi, Wen: PhD 2021, JINAN University, Associate prefessor

3. Admission information of HPGC

3.1 Programs and Costs

We can recruit graduate students from the following programs.

The Application fee is 800 RMB, and Accidental Injury and Hospitalization Insurance is 800 RMB /year. The tuition fees for different programs vary and are also listed in the table below

PhD/ Master	Code	Major	Remarks of Application	Tuition Fee of 2025
PhD	081200	Computer Science and Technology		40,000 RMB/Year
Master	0812J3	Data Science and Information Technology	This program takes English as the medium of instruction.	33,000 RMB/Year

Master	085404	Computer Technology		33,000 RMB/Year

3.2 Financial Aid

Scholarships will be granted on both merit and need basis.

3.2.1 Chinese Government Scholarship (CGS) - The Full CGS usually covers tuition, stipend, oncampus accommodation (or accommodation subsidy), and medical insurance. The Partial CGS includes one or a few items in the Full CGS. Annual Review is strictly required for all CGS holders.

3.2.2 Tuition Scholarship - Apply annually (**competitive**).

3.2.3 SIGS outstanding Scholarship - Applications from full-time international new students admitted to our institute will be ranked and selected based on their previous academic achievements, research outcomes, and overall performance. The specific amount of the scholarship will be tiered and distributed according to the results of the annual selection process.

3.2.4 Living allowance

SIGS provides living allowance for all the students: 33,000 RMB/year for doctor students and 7,200 RMB/year for master's students.

3.2.5 Assistantship

SIGS also offers different assistantship for outstanding students: teaching assistant and research assistant.

4. Contact

4.1 For HPGC Group related issues

Please contact Prof.Fu by mail haohuan@tsinghua.edu.cn or zhang.yating@sz.tsinghua.edu.cn with your CV.

4.2 For admission related issues

Please contact SIGS Admissions Office by mail

admission@sz.tsinghua.edu.cn or call +86 755 26036110

4.3 For scholarship related issues

Please contact SIGS Office of Academic Affairs by mail

pyc@sz.tsinghua.edu.cn or call +86 755 26036030

4.4 Contact of Programs

Code	Major	Contact
081200 / 085404	Computer Science and Technology / Computer Technology	Ms. Liu liurita@sz.tsinghua.edu.cn +86-755-86203272
0812J3	Data Science and Information Technology	Ms. Zhao datax@sz.tsinghua.edu.cn

4.5 For more information

Graduate Programs of Tsinghua University 2026 (For International Students):

https://yz.tsinghua.edu.cn/en/info/1035/1523.htm

4.6 2026 Application

The application closes at 17:00 p.m. on May 31, 2026 (Beijing Time).

Application Guide for Chinese Government Scholarship:

https://www.campuschina.org/content/details3_74776.html

4.7 Study in China

http://www.campuschina.org